Daffodil Institute of IT

Project Proposal of Final Semester - 2023
Department of Computer Science and Engineering
Daffodil Institute of IT (DIIT)

Project Title: **AI RESUME ANALYZER AND JOB MATCH SYSTEM**

Submitted by:

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## Objectives of the project:

To develop an intelligent system that analyzes resumes using AI techniques and matches candidates with suitable job opportunities based on their skills, experience, and qualifications.

## Limitations of the existing system:

- ❖ Manual resume screening
- ❖ High risk of human bias
- ❖ Static job matching

## Proposed System:

The proposed system is an intelligent AI-based platform designed to automate and enhance the recruitment process by analyzing resumes and matching candidates with suitable job opportunities. The system utilizes Natural Language Processing (NLP) and Machine Learning (ML) techniques to extract key information such as skills, education, experience, and certifications from uploaded resumes.

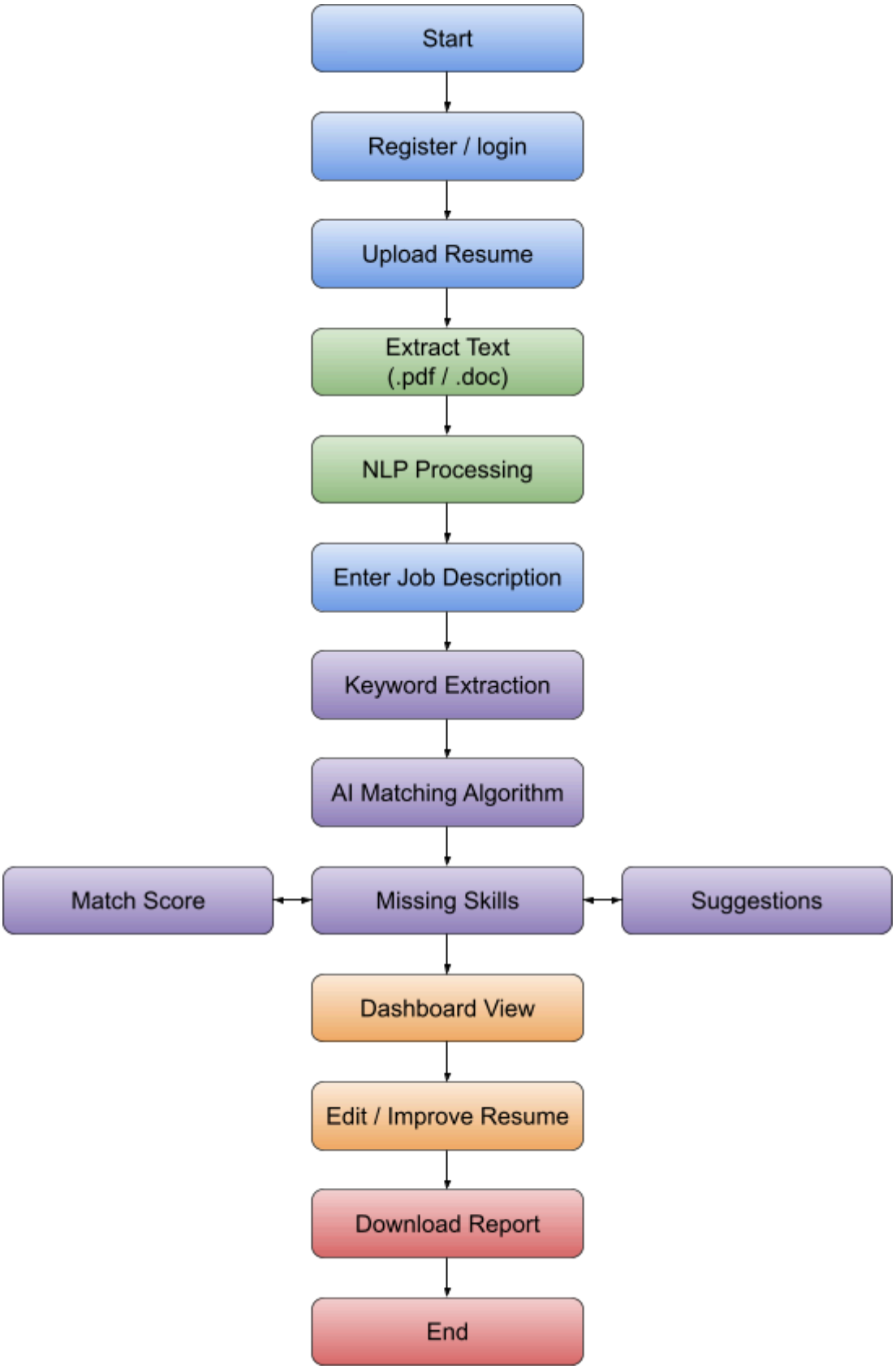
## Requirements:

- ❖ **Software:**
  - ➢ Operating System – Windows
  - ➢ Programming Language – Python
  - ➢ Framework – Django / Flask (Backend)
  - ➢ Frontend – HTML, CSS, JavaScript
  - ➢ Database – MySQL
  - ➢ Libraries – NLP libraries (NLTK, SpaCy)
  - ➢ API Tools – Postman / Swagger
  - ➢ Version Control – Git & GitHub
  - ➢ IDE – VS Code / PyCharm

## Features of the project:

- ❖ User uploads a resume – The user submits their resume file to the system.
- ❖ Text extraction – The system extracts readable text from PDF/DOC resumes.
- ❖ Skill, education, and experience identification – The system detects key information using NLP techniques.
- ❖ Job description input – The user provides a job description for comparison.
- ❖ AI comparison – The AI compares resume data with job requirements.
- ❖ Match score – The system generates a percentage score of compatibility.
- ❖ Missing skills – It identifies skills that are absent in the resume.
- ❖ Recommended improvements – The system suggests ways to improve the resume.
- ❖ Keyword analysis – It analyzes important keywords for optimization.
- ❖ REST API – The system provides API endpoints for communication between services.
- ❖ Authentication – It ensures secure user login and access control.
- ❖ File upload – It allows users to upload resume files safely.
- ❖ Validation – The system checks input data for accuracy and correctness.
- ❖ Database operations – It stores and retrieves user and resume data efficiently.
- ❖ AI inference – The system processes data using trained AI models to generate results.
- ❖ Swagger docs – It provides interactive API documentation for developers.
- ❖ Response modeling – It structures API responses in a consistent format.
- ❖ CV generation – The system automatically generates an improved CV based on analyzed data and suggestions.

## Workflow Diagram of the project:



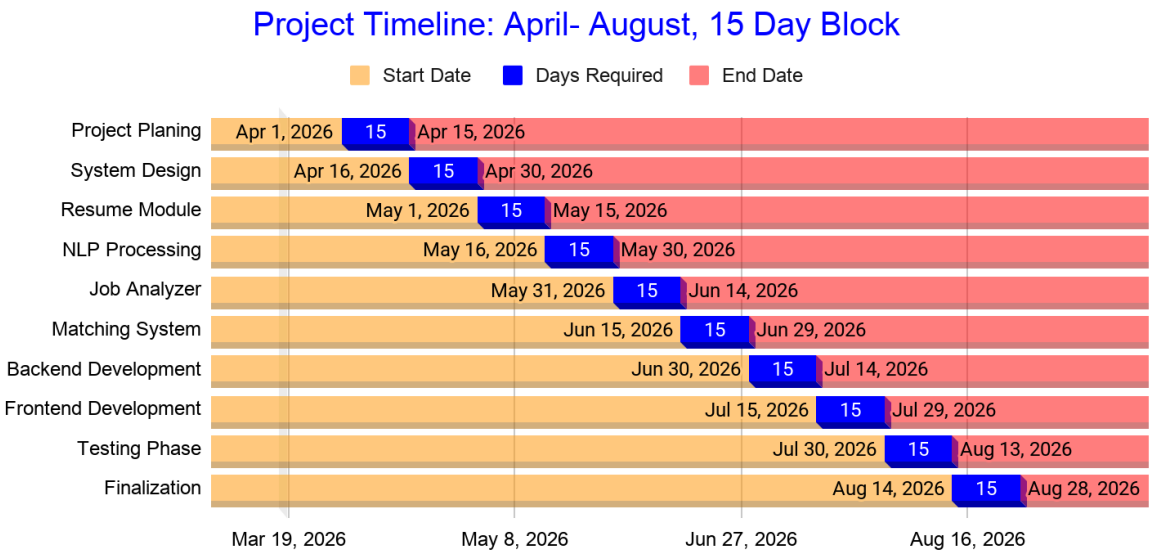
## Business Prospects:

- ❖ The system solves inefficient hiring by using AI to analyze resumes and match them with suitable jobs quickly.
- ❖ The target market includes HR companies, recruitment agencies, job portals, and university career services.
- ❖ The product offers resume analysis, skill extraction, job matching scores, and personalized improvement suggestions.
- ❖ Revenue is generated through subscription plans, premium features, and API integration services.
- ❖ The platform can scale globally with advanced AI features and integration with major recruitment systems in the future.

## Future Enhancements:

- ❖ Manual resume screening
- ❖ High risk of human bias
- ❖ Static job matching

## Gantt Chart:



## Declaration:

It is hereby declared that this project or any part of it has not been submitted elsewhere for the award of any degree.

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